

Region II of the Goodman–Style Odd-Cycle Bound: a corrected consolidated proof for $\frac{1}{2} < p < \frac{2}{3}$

Consolidated working note for T.K., H.Y., and collaborators

Operator and Huber reduction: GPT Pro, July 11, 2026;

scalar zone argument and certificates: Claude, July 11, 2026;

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Abstract

Let W be a graphon of edge density p , and let $m \geq 3$ be odd. This working note proves the remaining density range

$$\frac{1}{2} < p < \frac{2}{3}$$

of the proposed Goodman–style inequality

$$t(C_m, W) \geq p^m - p(1 - p)^{m-1}.$$

The proof has two logically separate parts. First, an operator-theoretic argument reduces every possible obstruction to one frontier eigenvalue α of the complement compression and then, through an exact Huber-type shape elimination, to the explicit scalar inequality

$$R_m \leq C_m \psi(\xi, \rho).$$

Second, that scalar inequality is proved on the full admissible parameter region. The pinch regime near $p = 1/2$, the small-frontier regime, and the Turán-corner sliver are handled analytically. Two compact intermediate regions are covered by finite, exhaustive certificates in exact rational arithmetic. Thus the final proof is rigorous but computer-assisted; the computer is used only for two finite semialgebraic/elementary-function covering checks, not for floating-point sampling.

This version repairs the supplied July 11 manuscript in four places: it corrects a reversed comparison in the m -maximisation lemma; states and proves the frontier-gap bound used by the Zone-C certifier; replaces several unsafe decimal roundings by directed rational bounds; and makes the logical status of the exact certificates explicit. The earlier proposed five-atom stop-loss argument is *not* used here: its scalar variables do not match the Huber interface, and no valid implication from that lemma to $R_m \leq C_m \psi(\xi, \rho)$ is currently known. The present document therefore contains only the proof whose bridge to the graphon problem has been fully checked.

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1 Introduction and main result

Let $W : [0, 1]^2 \rightarrow [0, 1]$ be a graphon with edge density $p = t(K_2, W)$, and let $m \geq 3$ be odd. The conjecture under study is

$$t(C_m, W) \geq g_m(p) := p^m - p(1-p)^{m-1}. \quad (1)$$

The consolidated record `earlier_consolidated_note.tex` (July 5, 2026) proves (1): for $p \leq 1/2$ (trivial, $g_m(p) \leq 0$); for every odd m when $p \geq 2/3$ (the high-density theorem); for all p and all odd $m \leq 13$ (per-cycle theorems C_3 – C_{13}); and for all p , all odd m , on several structured families (p -regular; positive-spectrum-safe; rank-one, rank-two and bipodal complements; equal cliques; two-value multipartite). The single remaining case is *Region II*:

$$\frac{1}{2} < p < \frac{2}{3}, \quad \text{i.e.} \quad q := 1 - p \in \left(\frac{1}{3}, \frac{1}{2}\right), \quad m \geq 15 \text{ odd.}$$

A July 11 working derivation by GPT Pro reduces Region II rigorously to a single *scalar* inequality — the *Huber payment inequality* — in the three variables (q, α, m) , where α is the unique eigenvalue of the complement compression above q (the “frontier eigenvalue”); all other spectral, eigenfunction-shape and coupling data are eliminated exactly. Section 2 restates that reduction, self-contained and with complete proofs, since everything that follows rests on it.

The present note proves the scalar inequality.

Theorem 1.1 (Scalar Huber inequality). *Let $q \in (\frac{1}{3}, \frac{1}{2})$, let $\alpha \in (q, r(q)]$ with $r(q) = \frac{\sqrt{q^2 + 4q} - q}{2}$, and let $m \geq 15$ be odd. With the notation of Definition 2.10 (all quantities explicit functions of (q, α, m)),*

$$R_m \leq C_m \psi(\xi, \rho).$$

Corollary 1.2 (Region II). *For every graphon W with $\frac{1}{2} < p < \frac{2}{3}$ and every odd $m \geq 3$,*

$$t(C_m, W) \geq p^m - p(1-p)^{m-1}.$$

Corollary 1.3 (The full conjecture). *For every graphon W and every odd $m \geq 3$, $t(C_m, W) \geq p^m - p(1-p)^{m-1}$, $p = t(K_2, W)$.*

Proof architecture. The scalar inequality is proved through the dual formula for the Huber envelope ψ . Zones A and B use $\lambda = 1$; Zone C uses either $\lambda = 1$ or $\lambda = 2\rho_0\xi$, where $0 < \rho_0 \leq \rho$ is an explicit lower bound and the latter choice is made only when it lies in $[0, 1]$. The admissible domain is covered by three zones: the pinch zone A ($e = 1 - 2\alpha \leq 1/60$, $\xi \geq 1$), zone B ($e \geq 1/60$, $\xi \geq 1$), and zone C ($\xi \leq 1$). Zone A and the small- e part of Zone C are elementary analytic estimates. Zone B is an analytic reduction to a two-variable inequality followed by a finite exact rational box certificate. The moderate- e part of Zone C is an analytic three-geometric reduction followed by a second finite exact rational certificate; its very thin Turán-corner remainder is again analytic.

Provenance and verification status

Result	Source	Status in this version
One-frontier and Huber reduction (§2)	GPT Pro, July 11	Pencil-and-paper proof; the endpoint issue in the forced-variance argument and the $K = 0$ shape case are stated explicitly.
Chart and scalar certificates (§3)	Claude, July 11	Pencil-and-paper identities; exact symbolic regression checks supplied.
Zone A (§4)	Claude, corrected here	Pencil-and-paper proof with all decimal constants interpreted as exact rationals and rounded in the safe direction.
Zone B (§5)	Claude, corrected here	Analytic m -maximisation plus the exact-rational verifier <code>zoneB.certifier.py</code> ; the reversed comparison in the manuscript is fixed.
Zone C (§6)	Claude, corrected here	Analytic for $e \leq 1/60$ and in the Turán corner; analytic reduction plus <code>zoneC.certifier.py</code> on the compact middle range. The missing bound $f \geq d + \delta$ is now proved and used.
Numerical sweep (§9)	Companion checker	Regression test only; no theorem depends on sampled floating-point values.

Remark 1.4 (Why the stop-loss note is not spliced into this proof). *A separate five-atom stop-loss inequality was derived during the audit. Its parameters use a different quantity $(q - \alpha^2)/(1 - \alpha)$, whereas the Huber reduction uses $L = \sqrt{pq - \alpha^2}$. The tempting domination that would bridge the two statements is false in general (already for admissible parameters at large m). Accordingly, that lemma is not invoked anywhere below. This remark is included to make the dependency graph unambiguous for later paper writing.*

2 The one-frontier reduction, recapped with proofs

Everything in this section is due to the July 11 baton note; we restate it so that the present document is self-contained, and we flag two small points of care (Remarks 2.5 and 2.6) that the original exposition leaves implicit.

2.1 Operator set-up and the Hilbert–Schmidt budget

Let $U = 1 - W$, let T_U be the integral operator with kernel U , and decompose $L^2([0, 1]) = \mathbb{R}\mathbf{1} \oplus \mathbf{1}^\perp$. In block form,

$$T_U = \begin{pmatrix} q & g^* \\ g & A \end{pmatrix}, \quad g = T_U\mathbf{1} - q\mathbf{1} \in \mathbf{1}^\perp, \quad A = P_{\mathbf{1}^\perp}T_UP_{\mathbf{1}^\perp}. \quad (2)$$

Since $0 \leq U \leq 1$ pointwise, $\int U^2 \leq \int U = q$, so $\|T_U\|_{\text{HS}}^2 \leq q$; expanding in blocks,

$$q^2 + 2\|g\|_2^2 + \text{Tr}(A^2) \leq q, \quad \text{i.e.} \quad \text{Tr}(A^2) + 2\|g\|_2^2 \leq pq. \quad (3)$$

2.2 The one-sided spectral shift

Proposition 2.1 (One-sided spectral shift). *For every odd $m \geq 3$,*

$$t(C_m, W) = p^m - \text{Tr}(A^m) + m[z^m]\mathcal{L}_W(z), \quad \mathcal{L}_W(z) = -\log\left(1 - \frac{z^2 \langle g, (I + zA)^{-1}g \rangle}{1 - pz}\right), \quad (4)$$

as an identity of formal power series coefficients.

Proof. For finite-rank step graphons, $t(C_m, W) = \text{Tr}(T_W^m)$ with $T_W = J - T_U$, $Jf = \langle f, \mathbf{1} \rangle \mathbf{1}$. Conjugating by the sign flip on $\mathbf{1}^\perp$ shows $\text{Tr}(T_W^m) = \text{Tr}(M_W^m)$ with $M_W = \begin{pmatrix} p & g^* \\ g & -A \end{pmatrix}$. Schur's determinant formula gives

$$\det(I - zM_W) = \det(I + zA) \left(1 - pz - z^2 \langle g, (I + zA)^{-1}g \rangle\right).$$

Taking $-\log$ and using $-\log \det(I - zT) = \sum_{r \geq 1} z^r \text{Tr}(T^r)/r$,

$$\sum_{r \geq 1} \frac{z^r}{r} \text{Tr}(M_W^r) = -\log \det(I + zA) - \log(1 - pz) + \mathcal{L}_W(z),$$

because $1 - pz - z^2 \langle g, (I + zA)^{-1}g \rangle = (1 - pz) \left(1 - \frac{z^2 \langle g, (I + zA)^{-1}g \rangle}{1 - pz}\right)$. Comparing the coefficient of z^m and multiplying by m yields (4), since $\text{Tr}((-A)^m) = -\text{Tr}(A^m)$ for odd m . For general graphons, let U_n be the conditional expectation of U on the dyadic grid of mesh 2^{-n} ; then $0 \leq U_n \leq 1$, $\int U_n = q$, and $U_n \rightarrow U$ in L^2 . Each side of (4) is continuous along this approximation: $t(C_m, \cdot)$ and $\text{Tr}(A^m)$ by telescoping and Schatten–Hölder ($m \geq 3 \geq 2$), and $m[z^m]\mathcal{L}_W$ because it is a polynomial in p and the finitely many moments $\langle g, A^j g \rangle$, $j \leq m - 2$, each continuous under $(g_n, A_n) \rightarrow (g, A)$ in $L^2 \times \text{HS}$. $\square$

Lemma 2.2 (Compression bound and the no-frontier case). *The compression A satisfies $\|A\| \leq \frac{1}{2}$. Moreover, if $\lambda_{\max}(A) \leq q$, then for every odd $m \geq 3$,*

$$t(C_m, W) \geq p^m - pq^{m-1}.$$

Proof. For a real $h \perp \mathbf{1}$, write $h = h^+ - h^-$ and $r = \int h^+ = \int h^- = \frac{1}{2}\|h\|_1$. Since $0 \leq U \leq 1$,

$$-2r^2 \leq \langle h, T_U h \rangle \leq 2r^2 \leq \frac{1}{2}\|h\|_2^2.$$

Taking the supremum over unit $h \in \mathbf{1}^\perp$ gives $\|A\| \leq 1/2 < p$.

As in the proof of Proposition 2.7, the formal series

$$u(z) = \frac{z^2 \langle g, (I + zA)^{-1}g \rangle}{1 - pz}$$

has nonnegative coefficients: for an eigenvalue $\nu \in [-1/2, 1/2]$ of A , the coefficient of z^n , $n \geq 2$, contributed by its coupling is

$$\frac{p^{n-1} - (-\nu)^{n-1}}{p + \nu} \geq 0.$$

Consequently $[z^m]\mathcal{L}_W(z) \geq 0$. If $\lambda_{\max}(A) \leq q$, then for every eigenvalue ν of A and odd $m \geq 3$,

$$\nu^m \leq q^{m-2}\nu^2.$$

Together with $\text{Tr}(A^2) \leq pq$, from (3), this yields $\text{Tr}(A^m) \leq pq^{m-1}$. Proposition 2.1 now gives the claim. $\square$

2.3 One frontier, forced variance, and the frontier ceiling

Fix $q \in (1/3, 1/2)$. By Lemma 2.2, only the case in which A has an eigenvalue $\alpha > q$ remains. Fix a corresponding unit eigenfunction $\phi \perp \mathbf{1}$.

Lemma 2.3 (One frontier). *A has at most one eigenvalue $> q$, and every other eigenvalue lies in $[-L, L]$, where*

$$L := \sqrt{pq - \alpha^2}, \quad 0 \leq L < q < \alpha < \frac{1}{2}.$$

Proof. Two eigenvalues $> q$ would give $\text{Tr}(A^2) > 2q^2 > q(1 - q) \geq \text{Tr}(A^2)$ for $q > 1/3$, by (3): contradiction. By (3) again, the total square mass of the non-frontier spectrum is at most $pq - \alpha^2 = L^2$, so each non-frontier eigenvalue has modulus at most L . Finally $L^2 = pq - \alpha^2 < pq - q^2 = q(1 - 2q) < q^2$ for $q > \frac{1}{3}$, so $L < q$; and $\alpha \leq \lambda_{\max}(A) \leq \|A\| \leq \|T_U\|_{\text{HS}} \leq \sqrt{q} < 1$, while $\alpha < \frac{1}{2}$ follows from Lemma 2.4 below. $\square$

Lemma 2.4 (Forced variance and the frontier ceiling). *Write $a = \int |\phi|$. Then $a^2 \geq 2\alpha$, and*

$$\|g\|_2^2 \geq \frac{\alpha(\alpha - q)^2}{2(1 - 2\alpha)}. \quad (5)$$

Consequently

$$\alpha \leq r(q) := \frac{\sqrt{q^2 + 4q} - q}{2}, \quad (6)$$

i.e. $\alpha^2 + q\alpha - q \leq 0$. (In particular $\alpha \leq r(q) < r(1/2) = 1/2$.)

Proof. Split $\phi = \phi^+ - \phi^-$. Since $U \geq 0$, $\alpha = \langle \phi, T_U \phi \rangle \leq \langle \phi^+, T_U \phi^+ \rangle + \langle \phi^-, T_U \phi^- \rangle$, and since $U \leq 1$, $\langle \phi^\pm, T_U \phi^\pm \rangle \leq (\int \phi^\pm)^2 = (a/2)^2$ (both signs have integral $a/2$ because $\int \phi = 0$). Hence $\alpha \leq a^2/2$.

Let $f = |\phi| = a\mathbf{1} + h$, $h \perp \mathbf{1}$, $\|h\|_2^2 = 1 - a^2$. Since $U \geq 0$, $4\langle \phi^+, T_U \phi^- \rangle = \langle f, T_U f \rangle - \langle \phi, T_U \phi \rangle \geq 0$, so $\langle f, T_U f \rangle \geq \alpha$. Expanding via (2) and $\langle h, Ah \rangle \leq \alpha\|h\|_2^2$,

$$\alpha \leq qa^2 + 2a\langle g, h \rangle + \alpha(1 - a^2) \leq qa^2 + 2a\|g\|_2\sqrt{1 - a^2} + \alpha(1 - a^2),$$

whence $(\alpha - q)a^2 \leq 2a\|g\|_2\sqrt{1 - a^2}$. Here $a^2 < 1$: if $a^2 = 1$, then the right-hand side vanishes while $\alpha > q$, a contradiction. Since $a^2 \geq 2\alpha$, this also gives $\alpha < 1/2$. We may therefore divide by $1 - a^2$ and obtain

$$\|g\|_2^2 \geq \frac{(\alpha - q)^2 a^2}{4(1 - a^2)} \geq \frac{\alpha(\alpha - q)^2}{2(1 - 2\alpha)},$$

using the monotonicity of $t \mapsto t/(1 - t)$. Combining (5) with $\alpha^2 + 2\|g\|_2^2 \leq pq$ (from (3), $\text{Tr}(A^2) \geq \alpha^2$) and clearing the positive denominator $1 - 2\alpha$ gives $(1 - \alpha - q)(\alpha^2 + q\alpha - q) \leq 0$; since $\alpha + q < 1$, the ceiling (6) follows. $\square$

Remark 2.5. *Two consequences of (3) are used repeatedly below: $\alpha^2 + 2\|g\|_2^2 \leq pq$, i.e. $\|g\|_2^2 \leq L^2/2$; and the non-frontier square mass is at most $L^2 - 2\|g\|_2^2 \geq 0$. The nonnegativity of the latter (automatic, by the former) is what makes the square-mass bookkeeping in Proposition 2.7 legitimate in all configurations.*

2.4 The master defect estimate

Decompose the degree fluctuation along the frontier:

$$g = c\phi + g_s, \quad c = \langle g, \phi \rangle, \quad g_s \perp \phi. \quad (7)$$

For odd m define on $(-p, p)$

$$k_m(\lambda) = \frac{p^{m-1} - \lambda^{m-1}}{p + \lambda}, \quad (8)$$

and set

$$A_m = 2L^{m-2} + m k_m(\alpha), \quad B_m = 2L^{m-2} + m k_m(L), \quad R_m = \alpha^m + L^m - pq^{m-1}. \quad (9)$$

Remark 2.6. k_m is positive on $(-p, p)$, strictly decreasing on $[0, p)$ (from $k'_m(\lambda)(p + \lambda) = -(m - 1)\lambda^{m-2} - k_m(\lambda)$), and for $\lambda \in [-L, 0]$ one has $k_m(\lambda) = \sum_{j=0}^{m-2} (-\lambda)^j p^{m-2-j} \geq p^{m-2} = k_m(0) \geq k_m(L)$. Hence $k_m(\lambda) \geq k_m(L)$ for all $\lambda \in [-L, L]$, and $0 < A_m < B_m$.

Proposition 2.7 (Master defect). *Suppose $\|g_s\|_2^2 \geq \tau$. Then*

$$t(C_m, W) - (p^m - pq^{m-1}) \geq -R_m + A_m c^2 + B_m \tau. \quad (10)$$

Proof. We bound the two non-baseline terms of (4).

(i) *The trace term.* By Lemma 2.3 and Remark 2.5, the non-frontier eigenvalues λ_i lie in $[-L, L]$ and satisfy $\sum_i \lambda_i^2 \leq L^2 - 2\|g\|_2^2$. For odd m ,

$$\text{Tr}(A^m) = \alpha^m + \sum_i \lambda_i^m \leq \alpha^m + \sum_{\lambda_i > 0} \lambda_i^m \leq \alpha^m + L^{m-2} \sum_{\lambda_i > 0} \lambda_i^2 \leq \alpha^m + L^m - 2L^{m-2}(c^2 + \|g_s\|_2^2).$$

(ii) *The shift term.* Write $u(z) = \frac{z^2 \langle g, (I + zA)^{-1} g \rangle}{1 - pz}$. Expanding g in the spectral resolution of A (eigenvalues $\nu_i \in [-L, L] \cup \{\alpha\}$, couplings c_i^2),

$$[z^n] u(z) = \sum_i c_i^2 \frac{p^{n-1} - (-\nu_i)^{n-1}}{p + \nu_i} \geq 0 \quad (n \geq 2),$$

since $|\nu_i| < p$. Thus u has nonnegative coefficients, hence so does $\mathcal{L}_W = \sum_{r \geq 1} u^r / r$, and

$$m[z^m] \mathcal{L}_W \geq m[z^m] u = m \sum_i c_i^2 k_m(\nu_i) \geq m k_m(\alpha) c^2 + m k_m(L) \|g_s\|_2^2,$$

using $(-\nu)^{m-1} = \nu^{m-1}$ ($m - 1$ even), Remark 2.6, and $\sum_{i \text{ safe}} c_i^2 = \|g_s\|_2^2$. Combining (i), (ii) with (4) and $\|g_s\|_2^2 \geq \tau$ gives (10). $\square$

2.5 Eigenfunction geometry: the two coupling channels

Set

$$z = \left(\int |\phi| \right)^2, \quad b = \langle |\phi|, \phi \rangle \quad (\geq 0 \text{ after } \phi \mapsto \pm \phi), \quad k = |\phi| - \sqrt{z} \mathbf{1} - b\phi, \quad K = \|k\|_2^2, \quad (11)$$

so that $z + b^2 + K = 1$ and, by Lemma 2.4, $z \geq 2\alpha$. Write $h = z - 2\alpha \geq 0$ and $e = 1 - 2\alpha > 0$, so $h + b^2 + K = e$.

Lemma 2.8 (Direct channel). $c \leq u_c := \frac{h - 2\alpha b}{2\sqrt{z}}$.

Proof. $T_U\phi = c\mathbf{1} + \alpha\phi$ by (2); also $T_U\phi \leq T_U\phi^+ \leq \int \phi^+ = \sqrt{z}/2$ pointwise ($U \leq 1, \phi \leq \phi^+$). Pairing with $\phi^+ \geq 0$, whose squared norm is $\langle \phi^+, \phi^+ \rangle = \frac{1+b}{2}$ (from $\|\phi^+\|^2 - \|\phi^-\|^2 = b, \|\phi^+\|^2 + \|\phi^-\|^2 = 1$):

$$\frac{c\sqrt{z}}{2} + \alpha \frac{1+b}{2} = \langle \phi^+, T_U\phi \rangle \leq \frac{\sqrt{z}}{2} \int \phi^+ = \frac{z}{4},$$

which rearranges to the claim using $z - 2\alpha = h$. $\square$

Lemma 2.9 (Safe channel). With $H = \frac{(\alpha - q)z + (\alpha - L)K}{2\sqrt{z}}$,

$$bc + \langle g_s, k \rangle \geq H, \quad \text{hence} \quad \|g_s\|_2^2 \geq \frac{(H - bc)_+^2}{K} \quad (K > 0).$$

Proof. $U \geq 0$ gives $\langle |\phi|, T_U|\phi| \rangle \geq \langle \phi, T_U\phi \rangle = \alpha$. Expanding the left side via $|\phi| = \sqrt{z}\mathbf{1} + b\phi + k$ and (2) ($\langle g, k \rangle = \langle g_s, k \rangle$ because $k \perp \phi; \langle \phi, Ak \rangle = 0$):

$$qz + 2\sqrt{z}(bc + \langle g_s, k \rangle) + \alpha b^2 + \langle k, Ak \rangle \geq \alpha = \alpha(z + b^2 + K).$$

Since $k \perp \mathbf{1}, \phi$ lies in the safe subspace, $\langle k, Ak \rangle \leq LK$ (Lemma 2.3); rearranging gives the first claim, and Cauchy–Schwarz $\langle g_s, k \rangle \leq \|g_s\|_2\sqrt{K}$ the second. $\square$

2.6 Exact Huber elimination

Definition 2.10. For admissible (q, α, m) put $d = \alpha - q, f = \alpha - L, e = 1 - 2\alpha$, and

$$C_m = \frac{B_m f \sqrt{2\alpha} e^2}{4\alpha^2}, \quad \xi = \frac{4\alpha^2 d}{e^2}, \quad \rho = \frac{A_m}{B_m} \frac{\sqrt{\alpha}}{2\sqrt{2}f}, \quad (12)$$

$$\psi(\xi, \rho) = \min_{0 \leq v \leq 1} \{ \rho v^2 + (\xi - v + v^2)_+ \}. \quad (13)$$

Theorem 2.11 (Exact Huber shape elimination; GPT Pro). For every Region-II graphon configuration with frontier data $(q, \alpha, c, g_s, \phi)$,

$$A_m c^2 + B_m \|g_s\|_2^2 \geq C_m \psi(\xi, \rho).$$

Consequently, by Proposition 2.7, Conjecture (1) holds in Region II as soon as the scalar inequality

$$R_m \leq C_m \psi(\xi, \rho) \quad (14)$$

holds for all $q \in (\frac{1}{3}, \frac{1}{2}), \alpha \in (q, r(q)]$ and odd $m \geq 15$.

Proof. Write $\mathcal{Q} = \min\{A_m c^2 + B_m K^{-1}(H - bc)_+^2 : c \leq u_c\}$ over the admissible shape data; by Lemmas 2.8 and 2.9 the left side dominates $\mathcal{Q}$ pointwise, so it suffices to prove $\mathcal{Q} \geq C_m \psi(\xi, \rho)$. (Degenerate case $K = 0$: then $k = 0$ and Lemma 2.9 gives $bc \geq H = d\sqrt{z}/2 \geq d\sqrt{2\alpha}/2 > 0$. Hence $c > 0$, so $u_c > 0$ and the budget (15) below applies. With $v = 2c\sqrt{2\alpha}/e$ it gives $b \leq \beta(c) \leq e(1 - v)/(2\alpha)$, and therefore

$$\frac{d\sqrt{2\alpha}}{2} \leq bc \leq c\beta(c) \leq \frac{e^2 v(1 - v)}{4\alpha\sqrt{2\alpha}}.$$

Equivalently $\xi \leq v - v^2$. Thus the positive-part term in (13) vanishes at this v , and $A_m c^2 = C_m \rho v^2 \geq C_m \psi$. We may henceforth assume $K > 0$.)

Case $u_c \leq 0$. Every admissible c is ≤ 0 ; since $b \geq 0$, $H - bc \geq H$, so $\mathcal{Q} \geq B_m H^2/K$. By AM-GM,

$$\frac{H^2}{K} = \frac{((\alpha - q)z + (\alpha - L)K)^2}{4zK} \geq (\alpha - q)(\alpha - L) = df,$$

and $C_m \psi \leq C_m \xi = B_m f \sqrt{2\alpha} d \leq B_m f d$ (take $v = 0$ in (13); $\sqrt{2\alpha} < 1$), so $\mathcal{Q} \geq C_m \psi$.

Case $u_c > 0$. A minimizer may be taken with $c \geq 0$: replacing $c < 0$ by 0 decreases both terms (for the second, $H - bc \geq H \geq H - b \cdot 0$ when $c \leq 0$, $b \geq 0$). The constraint $c \leq u_c$, i.e. $2c\sqrt{z} + 2ab \leq h = e - b^2 - K \leq e - b^2$, together with $z \geq 2\alpha$, gives

$$b^2 + 2ab + 2c\sqrt{2\alpha} \leq e, \quad (15)$$

so $0 \leq c \leq e/(2\sqrt{2\alpha})$ and $b \leq \beta(c) := \sqrt{\alpha^2 + e - 2c\sqrt{2\alpha}} - \alpha \leq \frac{e - 2c\sqrt{2\alpha}}{2\alpha}$. Also $z \leq 1$ and $z \geq 2\alpha$ give $H \geq \frac{d\sqrt{2\alpha}}{2} + \frac{f}{2}K$. For fixed c and any $K > 0$, AM-GM yields $\frac{(w+fK/2)^2}{K} \geq 2fw$ for $w \geq 0$, hence

$$\mathcal{Q} \geq \min_{0 \leq c \leq e/(2\sqrt{2\alpha})} \left\{ A_m c^2 + 2B_m f \left(\frac{d\sqrt{2\alpha}}{2} - c\beta(c) \right)_+ \right\}.$$

Substituting $v = 2c\sqrt{2\alpha}/e \in [0, 1]$ and using $c\beta(c) \leq \frac{e^2 v(1-v)}{4\alpha\sqrt{2\alpha}}$:

$$\frac{d\sqrt{2\alpha}}{2} - c\beta(c) \geq \frac{\sqrt{2\alpha} e^2}{8\alpha^2} (\xi - v + v^2), \quad A_m c^2 = \frac{A_m e^2}{8\alpha} v^2 = C_m \rho v^2,$$

and $2B_m f \cdot \frac{\sqrt{2\alpha} e^2}{8\alpha^2} = C_m$, so $\mathcal{Q} \geq C_m \min_{0 \leq v \leq 1} \{\rho v^2 + (\xi - v + v^2)_+\} = C_m \psi(\xi, \rho)$. $\square$

Proposition 2.12 (Forms of ψ ; GPT Pro). *Let $\xi_c(\rho) = \frac{2\rho+1}{4(\rho+1)^2}$ and, for $0 \leq \xi < \frac{1}{4}$, $v_-(\xi) = \frac{1-\sqrt{1-4\xi}}{2}$. Then*

$$\psi(\xi, \rho) = \begin{cases} \rho v_-(\xi)^2, & 0 \leq \xi < \xi_c(\rho), \\ \xi - \frac{1}{4(1+\rho)}, & \xi \geq \xi_c(\rho), \end{cases} \quad \text{and} \quad \psi(\xi, \rho) = \max_{0 \leq \lambda \leq 1} \left(\lambda \xi - \frac{\lambda^2}{4(\rho + \lambda)} \right).$$

Proof. On $\{v : \xi - v + v^2 \leq 0\}$ (nonempty iff $\xi \leq 1/4$) the objective is ρv^2 , minimized at $v_-(\xi)$; elsewhere it is $(\rho + 1)v^2 - v + \xi$ with unconstrained minimizer $v_* = \frac{1}{2(1+\rho)} \in (0, 1)$ and value $\xi - \frac{1}{4(1+\rho)}$; the branch transition $v_* = v_-(\xi)$ occurs exactly at $\xi = \xi_c(\rho)$. For the dual form, write $(t)_+ = \max_{0 \leq \lambda \leq 1} \lambda t$, exchange $\min_v \max_\lambda$ (convex-concave on compact intervals), and minimize $(\rho + \lambda)v^2 - \lambda v + \lambda \xi$ over $v \in [0, 1]$: the minimizer $\lambda/(2(\rho + \lambda)) \in [0, 1]$ gives the stated value. $\square$

The dual form furnishes the two *certificates* used throughout:

$$\lambda = 1 : \quad C_m \psi \geq C_m \left(\xi - \frac{1}{4(1+\rho)} \right) = \sqrt{2\alpha} B_m f \left(d - \frac{e^2}{16\alpha^2(1+\rho)} \right), \quad (16)$$

$$\lambda = 2\rho\xi (\leq 1) : \quad C_m \psi \geq C_m \rho \xi^2 \frac{1+4\xi}{1+2\xi} = \frac{2\alpha^3 A_m d^2}{e^2} \cdot \frac{1+4\xi}{1+2\xi}, \quad (17)$$

where we used $C_m \xi = \sqrt{2\alpha} B_m f d$ and $C_m \rho \xi^2 = 2\alpha^3 A_m d^2/e^2$ (both exact, from (12)).

3 The scalar problem in frontier-gap coordinates

3.1 The admissible domain and the (e, κ) chart

Definition 3.1 (Admissible domain). $\mathcal{D}$ is the set of triples (q, α, m) with $q \in (\frac{1}{3}, \frac{1}{2})$, $\alpha \in (q, r(q)]$, $m \geq 15$ odd. Throughout, $p = 1 - q$, $d = \alpha - q$, $e = 1 - 2\alpha$, $L = \sqrt{pq - \alpha^2}$, $f = \alpha - L$, and we use the ratios

$$x = \frac{\alpha}{p}, \quad \tau = \frac{q}{\alpha}, \quad y = \frac{L}{p}, \quad \ell = \frac{L}{\alpha}, \quad u = 1 - x = \frac{d+e}{p}, \quad T = me, \quad \kappa = \frac{d}{e}.$$

Lemma 3.2 (The (e, κ) chart). The map $(q, \alpha) \mapsto (e, \kappa)$ identifies $\{(q, \alpha) : q \in (\frac{1}{3}, \frac{1}{2}), \alpha \in (q, r(q))\}$ with

$$\left\{ (e, \kappa) : 0 < e < \frac{1}{3}, \quad 0 < \kappa \leq \kappa_{\max}(e) := \frac{1-e}{1+e}, \quad \kappa < \kappa_q(e) := \frac{1-3e}{6e} \right\},$$

via $\alpha = \frac{1-e}{2}$, $q = \alpha - \kappa e$, $p = \alpha + (1 + \kappa)e$. On this chart,

$$L^2 = \alpha e - d(d+e), \quad \xi = \frac{(1-e)^2 \kappa}{e}, \quad \frac{1}{x} = 1 + \frac{2(1+\kappa)e}{1-e}, \quad 1 - \tau = \frac{2\kappa e}{1-e}.$$

Proof. $e = 1 - 2\alpha \in (0, \frac{1}{3})$ iff $\alpha \in (\frac{1}{3}, \frac{1}{2})$, which holds since $q > \frac{1}{3}$ and $\alpha \leq r(q) < r(\frac{1}{2}) = \frac{1}{2}$. Next, $q > \frac{1}{3}$ iff $\kappa e < \alpha - \frac{1}{3} = \frac{1-3e}{6}$, i.e. $\kappa < \kappa_q(e)$. For the ceiling: a direct computation gives

$$\alpha^2 + q\alpha - q = \alpha^2 + q(\alpha - 1) = \frac{(1-e)^2}{4} - \left(\frac{1-e}{2} - \kappa e \right) \frac{1+e}{2} = \frac{e}{2} (\kappa(1+e) - (1-e)),$$

so $\alpha \leq r(q)$ (i.e. $\alpha^2 + q\alpha - q \leq 0$) iff $\kappa \leq \kappa_{\max}(e)$. Also $L^2 = pq - \alpha^2 = (\alpha + d + e)(\alpha - d) - \alpha^2 = \alpha e - d(d+e)$, $\xi = 4\alpha^2 d / e^2 = (1-e)^2 \kappa / e$, and the last two identities are immediate from $p = \alpha + (1+\kappa)e$, $q = \alpha - \kappa e$, $\alpha = \frac{1-e}{2}$. $\square$

Lemma 3.3 (Elementary facts on $\mathcal{D}$). On $\mathcal{D}$: $0 \leq L < q < \alpha < \frac{1}{2} < p$; $y < \frac{1}{2}$; $\ell \leq \sqrt{e/\alpha} = \sqrt{2e/(1-e)}$; and for every integer $n \geq 2$ and $0 < u < 1$,

$$(1-u)^n \leq \frac{1}{1+nu + \binom{n}{2}u^2}, \quad (1-u)^n \geq 1 - nu. \quad (18)$$

Moreover the defect has the exact form

$$R_m = \alpha^{m-1}(\alpha - p\tau^{m-1}) + L^m. \quad (19)$$

Proof. $L < q$: $L^2 = pq - \alpha^2 < pq - q^2 = q(1-2q) < q^2$ for $q > \frac{1}{3}$. $y < \frac{1}{2}$: with $s = q/p \in (\frac{1}{2}, 1)$ and $x > s$ one has $y^2 = pq/p^2 - x^2 = s - x^2 < s - s^2 \leq \frac{1}{4}$. $\ell^2 = L^2/\alpha^2 \leq \alpha e/\alpha^2 = e/\alpha$. (18): the binomial series $(1-u)^{-n} = \sum_k \binom{n+k-1}{k} u^k$ has nonnegative terms, so $(1-u)^{-n} \geq 1 + nu + \binom{n+1}{2}u^2 \geq 1 + nu + \binom{n}{2}u^2$; Bernoulli is standard. (19): $q = \alpha\tau$ and $\alpha^m + L^m - pq^{m-1} = \alpha^{m-1}\alpha - p\alpha^{m-1}\tau^{m-1} + L^m$. $\square$

Lemma 3.4 (Frontier-gap inequalities). Put

$$\delta := \alpha - \frac{1}{3} > 0.$$

Then, throughout the admissible domain,

$$d + L \leq \frac{1}{3}, \quad q - L \geq \delta, \quad f = \alpha - L \geq d + \delta. \quad (20)$$

Proof. Since $q = \alpha - d = \frac{1}{3} + \delta - d$ and $L^2 = pq - \alpha^2$, direct expansion gives

$$\left(\frac{1}{3} - d\right)^2 - L^2 = 2(d^2 - d\delta + \delta^2) + \frac{\delta - d}{3} \geq 0.$$

The domain condition $q > 1/3$ is exactly $0 < d < \delta$, so $1/3 - d > 0$. Hence $L \leq 1/3 - d$, proving the first inequality. The other two follow from

$$q - L = \alpha - (d + L) \geq \alpha - \frac{1}{3} = \delta, \quad f = d + (q - L).$$

□

3.2 The three zones

Theorem 1.1 is proved by combining three lemmas whose hypotheses cover $\mathcal{D}$ (recall the two certificates (16), (17)):

- **Zone A** (§4): $e \leq \frac{1}{60}$ and $\xi \geq 1$; certificate $\lambda = 1$. This zone contains the *pinch* $q \uparrow \frac{1}{2}$, where the inequality is asymptotically tight (relative margin $\Theta(1/m)$); its proof is the sharpest.
- **Zone B** (§5): $e \geq \frac{1}{60}$ and $\xi \geq 1$; certificate $\lambda = 1$. (Nonempty only for $e < 0.2033$: $\xi \geq 1$ requires $\kappa \geq e/(1-e)^2$, while the domain requires $\kappa < \kappa_q(e)$; the two are compatible iff $6e^2 < (1-e)^2(1-3e)$, whose left-minus-right side is strictly increasing in e and changes sign in $(0.2032, 2033/10000)$ — exact rational evaluation at the endpoints.)
- **Zone C** (§6): $\xi \leq 1$ (any e); the admissible dual choice is $\lambda = 2\rho_0\xi$ when this is at most 1, and $\lambda = 1$ otherwise, for an explicit $\rho_0 \leq \rho$. This zone contains the ultra-thin frontier scale and the *Turán corner* $e \uparrow \frac{1}{3}$, $\kappa \downarrow 0$, where $R_m \rightarrow 0$ and the constraint $q > \frac{1}{3}$ is essential.

4 Zone A: $e \leq 1/60$, $\xi \geq 1$

Lemma 4.1 (Zone A). *Let $(q, \alpha, m) \in \mathcal{D}$ with $e \leq \frac{1}{60}$ and $\xi \geq 1$. Then*

$$R_m \leq \sqrt{2\alpha} B_m f\left(d - \frac{e^2}{16\alpha^2(1+\rho)}\right) \leq C_m \psi(\xi, \rho). \quad (21)$$

The second inequality is the dual certificate (16); the content is the first. We may assume $R_m > 0$. All terminating decimals in this section denote the displayed exact rational numbers. Put

$$z := (m-2)u, \quad \varepsilon := \frac{e^2}{16\alpha^2(1+\rho)d} = \frac{e}{4(1-e)^2(1+\rho)\kappa}, \quad \sigma_e := \left(\frac{51}{25}e\right)^6.$$

For $e \leq 1/60$ and $0 < \kappa \leq 1$ we shall use

$$\alpha \geq \frac{59}{120}, \quad p \leq \frac{1+3e}{2} \leq \frac{21}{40}, \quad x \geq \frac{1-e}{1+3e} \geq 0.9365. \quad (22)$$

$$\frac{e}{\alpha} = \frac{2e}{1-e} \leq 2.04e, \quad \ell \leq \sqrt{\frac{2e}{1-e}} \leq 1.4262\sqrt{e} \leq 0.1842. \quad (23)$$

The last two decimal comparisons follow, for example, from $120/59 < 2.034$ and $(1.4262)^2 > 120/59$.

Lemma 4.2 (Positivity threshold in Zone A). *If $R_m > 0$, then*

$$\kappa(m-1) > x(1+\kappa) - \sigma_e, \quad \kappa \geq \frac{0.93}{m-1}. \quad (24)$$

Proof. From (19), $R_m > 0$ gives $\tau^{m-1} < x + L^m/(p\alpha^{m-1})$. Since $L^2 \leq \alpha e$ and $m \geq 15$,

$$\frac{L^m}{p\alpha^{m-1}} \leq x \left(\frac{e}{\alpha}\right)^{m/2} \leq x(2.04e)^{7.5} = x\sigma_e(2.04e)^{1.5}.$$

Bernoulli's inequality gives $\tau^{m-1} \geq 1 - (m-1)d/\alpha$, and hence

$$(m-1)\frac{d}{\alpha} > 1 - x(1 + \sigma_e(2.04e)^{1.5}) \geq \frac{d+e}{p} - \sigma_e(2.04e)^{1.5}.$$

After multiplication by α/e , the first term becomes $x(1+\kappa)$, whereas $\alpha 2.04^{1.5}\sqrt{e} < 0.19$. This proves the first inequality. Since $x - \sigma_e > 0.93$, it also gives the second. $\square$

Lemma 4.3 (Payment and defect reduction). *Under the hypotheses above,*

$$\varepsilon \leq \min\left(\frac{1}{4}, 0.2781 T\right), \quad T := me, \quad (25)$$

and (21) follows from

$$\sqrt{1-e} \frac{(1-\ell)(1-y^{m-1})(1-\varepsilon)}{1+y} \geq x^{m-2} \left[\frac{m-1}{mx} - \frac{1+1/\kappa}{m} \right] + \Lambda, \quad \Lambda := \frac{(\alpha e)^{m/2}}{md\alpha p^{m-2}}. \quad (26)$$

Moreover $\Lambda \leq \sigma_e x^{m-2}$.

Proof. The condition $\xi \geq 1$ gives $d \geq e^2/(4\alpha^2)$, hence $\varepsilon \leq 1/[4(1+\rho)] \leq 1/4$. Lemma 4.2 also gives

$$\varepsilon \leq \frac{(m-1)e}{0.93 \cdot 4(1-e)^2} < 0.2781 me,$$

because $[0.93 \cdot 4(59/60)^2]^{-1} < 0.2781$.

Next,

$$B_m \geq mk_m(L) = \frac{mp^{m-1}(1-y^{m-1})}{p+L}.$$

Using $\sqrt{2\alpha} = \sqrt{1-e}$, $f = \alpha(1-\ell)$ and $p+L = p(1+y)$ gives the left side of (26), multiplied by $md\alpha p^{m-2}$. On the other hand, Bernoulli's inequality in (19) gives

$$R_m \leq \alpha^{m-1} \left[\frac{(m-1)d}{x} - (d+e) \right] + (\alpha e)^{m/2}.$$

Division by $md\alpha p^{m-2}$ yields (26). Finally,

$$\Lambda = x^{m-2} \frac{(e/\alpha)^{m/2-1}}{m\kappa} \leq x^{m-2} \frac{(2.04e)^{6.5}}{0.93} \leq \sigma_e x^{m-2},$$

since $m\kappa \geq 0.93$ and $\sqrt{2.04e} < 0.19 < 0.93$. $\square$

Proof of Lemma 4.1. Let the left and right sides of (26) be LHS and RHS. Since $y \leq \ell$, $(1 - \ell)/(1 + \ell) \geq 1 - 2\ell$, $\sqrt{1 - e} \geq 1 - 0.51e$, and $y^{m-1} \leq \ell^{14} \leq (2.04e)^7 \leq \sigma_e$, we have

$$\text{LHS} \geq 1 - \Gamma, \quad \Gamma := 2\ell + \varepsilon + 0.51e + \sigma_e. \quad (27)$$

If the square bracket in (26) is negative, then $\text{RHS} \leq \Lambda \leq \sigma_e$, whereas $\text{LHS} > 0.36$, and there is nothing to prove. Assume from now on that the bracket is nonnegative.

We have

$$[\dots] \leq \frac{1}{x} - \frac{2 + 1/\kappa}{m}, \quad \frac{1}{x} \leq 1 + 2.034(1 + \kappa)e,$$

and, by (18),

$$x^{m-2} \leq \frac{1}{1 + z + \frac{6}{13}z^2}.$$

Furthermore

$$u = \frac{2(1 + \kappa)e}{1 + e + 2\kappa e} \geq \frac{40}{21}(1 + \kappa)e > 1.9047(1 + \kappa)e.$$

Consequently

$$2.034(1 + \kappa)e \leq 0.0822z, \quad (28)$$

$$T \leq 0.6058z, \quad \varepsilon \leq 0.1686z, \quad (29)$$

$$\frac{2 + 1/\kappa}{m} \geq \frac{11.08e}{z + 7.8e}. \quad (30)$$

For (30), use $(1 + \kappa)(2 + 1/\kappa) \geq 3 + 2\sqrt{2} > 5.8284$ and $1 + \kappa \leq 2$.

Thus (26) follows from

$$(1 - \Gamma) \left(1 + z + \frac{6}{13}z^2 \right) \geq 1 + 0.0822z - \frac{11.08e}{z + 7.8e} + \sigma_e, \quad (31)$$

where

$$\Gamma \leq A_0 + \min \left(\frac{1}{4}, 0.1686z \right), \quad A_0 := 2.8527\sqrt{e} + 0.51e + \sigma_e.$$

The term σ_e on the right absorbs Λ because $\Lambda \leq \sigma_e x^{m-2}$ and $x^{m-2}(1 + z + \frac{6}{13}z^2) \leq 1$.

Let F denote the left side of (31) minus the right side. Then

$$F = z(0.9178 - \Gamma) + \frac{6}{13}(1 - \Gamma)z^2 + \frac{11.08e}{z + 7.8e} - \Gamma - \sigma_e.$$

For $e \leq 1/60$,

$$A_0 \leq 0.3768, \quad \sigma_e \leq 10^{-7}e. \quad (32)$$

We check three ranges.

Range 1: $0 < z \leq 1$. Using $\Gamma \leq A_0 + 0.1686z$, and then $z \leq 1$, gives

$$F \geq 0.3724z + \frac{11.08e}{z + 7.8e} - A_0 - \sigma_e;$$

the discarded z^2 coefficient is at least 0.041. By AM–GM,

$$0.3724z + \frac{11.08e}{z + 7.8e} \geq 4.0626\sqrt{e} - 2.9048e.$$

Therefore

$$F \geq 1.2099\sqrt{e} - 3.4148e - 2\sigma_e > 0,$$

since $3.4148\sqrt{e} < 0.441$ and (32) holds.

Range 2: $1 \leq z \leq 1.4827$. Here $0.1686z < 1/4$. Substitution of $A_0 \leq 0.3768$ gives

$$F \geq q(z) := 0.3724z + 0.1190z^2 - 0.0779z^3 - 0.3768 - \sigma_e.$$

The derivative $q'(z) = 0.3724 + 0.2380z - 0.2337z^2$ is positive on this interval (it is concave and positive at both endpoints), while $q(1) > 0.036 - \sigma_e > 0$.

Range 3: $z \geq 1.4827$. Now $\Gamma \leq A_0 + 1/4 \leq 0.6268$, so

$$F \geq 0.2910z + 0.1722z^2 - 0.6268 - \sigma_e.$$

The right side is increasing and exceeds $0.183 - \sigma_e$ at $z = 1.4827$. Thus $F > 0$ in every range, proving (31) and the lemma. $\square$

Remark 4.4 (Where Zone A is tight). *In the pinch corner ($e \rightarrow 0$, $T = me \rightarrow 0$, κ fixed), payment and defect agree to leading order: both are $2\kappa me p^m(1+o(1))$, and the margin is $(4\kappa+2)e p^m(1+o(1))$, i.e. a relative margin $\sim (2 + 1/\kappa)/m$. Thus the relative margin can tend to zero on scales of order $1/m$. In particular, no proof with a uniform positive relative margin is possible; Range 1 above must retain the $O(\sqrt{e})$ and $O(1/m)$ effects rather than discarding them. The optional numerical scan in §9 is consistent with this asymptotic picture.*

5 Zone B: $e \geq 1/60$, $\xi \geq 1$

Lemma 5.1 (Zone B). *Let $(q, \alpha, m) \in \mathcal{D}$ with $\frac{1}{60} \leq e < \frac{1}{3}$, $\xi \geq 1$ and $R_m > 0$. Then $R_m \leq \sqrt{2\alpha} B_m f(d - \frac{e^2}{16\alpha^2(1+\rho)}) \leq C_m \psi(\xi, \rho)$.*

As noted in §3, the hypotheses force $e < \frac{2033}{10000}$ (else $\kappa_\xi \geq \kappa_q$ and the zone is empty). The payment and defect reductions of Lemma 4.3 are domain-free, so, exactly as there, it suffices to prove the reduced inequality

$$\Pi := \sqrt{1-e} \frac{(1-\ell)(1-y^{m-1})(1-\varepsilon)}{1+y} \geq x^{m-2} \left[\frac{m-1}{mx} - \frac{1+1/\kappa}{m} \right] + \Lambda. \quad (33)$$

The new difficulty is that the worst cycle length m is now an interior maximum (roughly $m \approx 1 + A + 1/\lambda$ below), so we first organise the right side as a one-variable maximization.

Lemma 5.2 (The m -maximization). *Write $n = m - 1 \geq 14$, $A = x(1 + \kappa)/\kappa$, $\lambda = \ln(1/x)$, and $K(n) = x^n(n - A)/(n + 1)$, so that $x^{m-2} \left[\frac{m-1}{mx} - \frac{1+1/\kappa}{m} \right] = K(n)/x^2$. Then:*

- (i) *If $n \leq A$ the bracket is ≤ 0 and (33) reduces to $\Pi \geq \Lambda$.*
- (ii) *K is strictly log-concave on (A, ∞) , with stationary point $n_* = A + s_*$, where $s_*(s_* + A + 1) = (A + 1)/\lambda$. Hence $\max_{n \geq 14} K(n) = K(\max(14, n_*))$.*
- (iii) *$K(14) \leq x^{14}(14 - A)_+/15$, and if $A < 14$ then $n_* \leq 14$ if and only if $15(14 - A)\lambda \geq A + 1$.*

(iv) With $\nu := \lambda(A+1)$ and $w := (\sqrt{\nu^2 + 4\nu} - \nu)/2$,

$$K(n_*) = e^{-\lambda A} e^{-w}(1-w) \leq e^{-\lambda A - 2w},$$

and the map $(a, \nu) \mapsto e^{-a-2w(\nu)}$ is decreasing in both arguments; moreover

$$\lambda A \geq \nu' \quad \text{and} \quad \nu \geq \nu', \quad \nu' := \Phi(e, \kappa) := \frac{2(1-e)(1+\kappa)^2 e}{\kappa(1+e+2\kappa e)^2},$$

so $K(n_*) \leq e^{-\Phi-2w(\Phi)}$.

Proof. (i) is clear ($\Lambda \geq$ the whole right side then). (ii): $(\ln K)'' = -\frac{1}{(n-A)^2} + \frac{1}{(n+1)^2} < 0$ since $0 < n-A < n+1$; the stationarity condition $\lambda = \frac{1}{n-A} - \frac{1}{n+1} = \frac{A+1}{(n-A)(n+1)}$ gives the quadratic for $s_* = n_* - A$; unimodality gives the max formula. (iii): the first claim is $(n-A)/(n+1)$ at $n=14$; for the second, $s \mapsto s(s+A+1)$ is increasing, so

$$s_* \leq 14-A \iff \frac{A+1}{\lambda} = s_*(s_*+A+1) \leq 15(14-A),$$

which is equivalent to $15(14-A)\lambda \geq A+1$. (iv): from $s_*(s_*+A+1) = (A+1)/\lambda$, multiplying by λ^2 gives $w^2 + \nu w = \nu$ with $w = \lambda s_*$, so $w = (\sqrt{\nu^2 + 4\nu} - \nu)/2 < 1$ and $\frac{s_*}{s_*+A+1} = \frac{\lambda s_*^2}{(A+1)/\lambda} = \frac{w^2}{\nu} = 1-w$; hence $K(n_*) = x^{A+s_*} \frac{s_*}{s_*+A+1} = e^{-\lambda A} e^{-w}(1-w) \leq e^{-\lambda A - 2w}$ by $1-w \leq e^{-w}$. Monotonicity in a is clear; in ν , $w'(\nu) = \frac{1}{2}(\frac{\nu+2}{\sqrt{\nu^2+4\nu}} - 1) > 0$ since $(\nu+2)^2 > \nu^2 + 4\nu$. Finally $\lambda A \geq uA = \frac{2(1+\kappa)e}{1+e+2\kappa e} \cdot \frac{(1-e)(1+\kappa)}{\kappa(1+e+2\kappa e)} = \Phi$ using $\lambda \geq u = 1-x$ and the chart formulas, and $\nu = \lambda(A+1) \geq \lambda A \geq \Phi$. $\square$

Lemma 5.3 (Zone-B battle). *For all $e \in [\frac{1}{60}, \frac{2033}{10000}]$ and $\kappa \in [\kappa_\xi(e), \min(\kappa_{\max}, \kappa_q)(e)]$:*

$$\underline{\Pi}(e, \kappa) \geq \frac{\mathfrak{K}(e, \kappa)}{x^2} + \bar{\Lambda}(e), \quad (34)$$

where $\underline{\Pi} = \sqrt{1-e} \frac{(1-\bar{\ell})(1-\bar{\ell}^{14})(1-\bar{\varepsilon})}{1+\bar{\ell}}$ with $\bar{\ell} = \sqrt{2e/(1-e)}$, $\bar{\varepsilon} = \min(\frac{1}{4}, \frac{e}{4(1-e)^2\kappa(1+\rho)})$, $\rho = \frac{1}{4}(1 - e^{-17.37(1+\kappa)e})$; $\mathfrak{K} = x^{14}(14-A)_+/15$ when the gate $15(14-A)\lambda \geq A+1$, $A < 14$ holds, and $\mathfrak{K} = \max(x^{14}(14-A)_+/15, e^{-\Phi-2w(\Phi)})$ otherwise; and $\bar{\Lambda} = x^{13}(2e/(1-e))^{6.5}(1-e)^2/(15e)$.

Proof. This is a two-variable inequality between explicit elementary functions; it is certified by exhaustive subdivision of the strip into rational boxes with monotone endpoint bounds, in exact rational arithmetic (integer-square-root bracketing for $\sqrt{\cdot}$, partial-sum bracketing with tail bounds for e^{-t}): script `zoneB_certifier.py`, which verifies the cover and exits successfully. The certificate is small: 23 verified boxes (3 further boxes lie outside the admissible strip), maximum subdivision depth 8; each box is checkable by hand from the monotone endpoint bounds below. The monotonicity facts used for the box bounds are: x is decreasing in e and κ ; $(1+\kappa)^2/\kappa$ is decreasing for $\kappa \leq 1$; $(1-e)e$ is increasing for $e \leq \frac{1}{2}$; κ_ξ is increasing and $\kappa_{\max}, \kappa_q$ decreasing in e ; $w(\cdot)$ is increasing; e^{-t} decreasing. $\square$

Proof of Lemma 5.1. The hypotheses give $\kappa \geq \kappa_\xi(e)$ ($\xi \geq 1$) and $\kappa \leq \min(\kappa_{\max}, \kappa_q)$ (Lemma 3.2); as computed above the strip is empty unless $e < \frac{2033}{10000}$. The quantities in (33) obey $\Pi \geq \underline{\Pi}$ (using $\ell \leq \bar{\ell}$, $y \leq \ell$, $y^{m-1} \leq \bar{\ell}^{14}$, $\varepsilon \leq \bar{\varepsilon}$, the latter by $\rho \geq (1-x^{m-1})\frac{\sqrt{\alpha}}{2\sqrt{2}f(1+x)} \geq \frac{1}{4}(1-e^{-14u}) \geq \rho$). Here the first

inequality uses $A_m/B_m \geq k_m(\alpha)/k_m(L) \geq (1 - x^{m-1})/(1 + x)$, and $\frac{\sqrt{\alpha}}{2\sqrt{2}f(1+x)} \geq \frac{1}{2\sqrt{2}\sqrt{\alpha}(1+x)} \geq \frac{1}{4}$ and, because $e \leq 2033/10000$,

$$14u \geq \frac{28(1 + \kappa)e}{1 + 3e} \geq 17.37(1 + \kappa)e$$

(the last comparison is an exact rational inequality)); $\Lambda \leq \bar{\Lambda}$ (the ratio of consecutive odd- m values of Λ is $x^2(e/\alpha) \cdot \frac{m}{m+2} < 1$, so the $m = 15$ value bounds all, and $\kappa \geq \kappa_\xi$ bounds $1/(15\kappa)$); and by Lemma 5.2 the right side of (33) is at most $\mathfrak{R}/x^2 + \Lambda$. Now Lemma 5.3 closes (33), hence the certificate inequality, hence the lemma by (16). $\square$

6 Zone C: $\xi \leq 1$

Lemma 6.1 (Zone C). *Let $(q, \alpha, m) \in \mathcal{D}$ with $\xi \leq 1$ and $R_m > 0$. Then $R_m \leq C_m \psi(\xi, \rho)$. More precisely, the proof exhibits an admissible value of the dual parameter: in the small- e range it uses $\lambda = 2\rho\xi < 1$; in the moderate- e range it uses either $\lambda = 2\rho_{10}\xi \leq 1$ or $\lambda = 1$, where the explicit lower bound $\rho_{10} \leq \rho$ is defined in Lemma 6.4.*

We prove the small- e half here (Lemma 6.2); the moderate- e half, which contains the Turán corner, is Lemma 6.3 in the next subsection. Note $\xi \leq 1$ is equivalent to $\kappa \leq \kappa_\xi := e/(1 - e)^2$, i.e. $d \leq e^2/(1 - e)^2$: the frontier sits very close to q .

6.1 Small e

Lemma 6.2 (Zone C, $e \leq 1/60$). *Lemma 6.1 holds when $e \leq \frac{1}{60}$.*

Proof. Lemma 4.2 applies, because its proof used only $e \leq 1/60$. Thus $\kappa \geq 0.93/(m - 1)$. Since $\kappa \leq \kappa_\xi = e/(1 - e)^2$,

$$m \geq 55, \quad T = me > (m - 1)e \geq 0.93(1 - e)^2 \geq 0.899. \quad (35)$$

The branch $2\rho\xi > 1$ is empty. Because $k_m(\alpha) \leq k_m(L)$,

$$\frac{A_m}{B_m} \leq \frac{2y^{m-2}(1 + y)}{m(1 - y^{m-1})} + \frac{1 + y}{1 + x} \leq 10^{-38} + 0.6117.$$

Here we used $y \leq \ell \leq 0.1842$, $x \geq 0.9365$, and $m \geq 55$. Moreover $f = \alpha(1 - \ell) \geq 0.8157\alpha$, so

$$\frac{\sqrt{\alpha}}{2\sqrt{2}f} \leq 0.6182.$$

Consequently $\rho \leq 0.3782$ and $2\rho\xi \leq 0.7564 < 1$.

Since $2\rho\xi < 1$, the dual choice $\lambda = 2\rho\xi$ gives the value in (17). It is therefore enough to prove the slightly stronger inequality

$$R_m \leq 2\alpha^3 A_m \kappa^2. \quad (36)$$

Since

$$(m - 1)u \geq 1.9047(m - 1)e \geq 1.712, \quad 1 - x^{m-1} \geq 1 - e^{-1.712} > 0.8142,$$

we have

$$A_m \geq mk_m(\alpha) = \frac{mp^{m-2}(1 - x^{m-1})}{1 + x}.$$

Put

$$t := (m-1)\log(1/\tau) - \log(1/x), \quad z := \kappa m.$$

Then $\alpha - p\tau^{m-1} = \alpha(1 - e^{-t})$ and

$$R_m \leq \alpha^m \min(1, t_+) + L^m, \quad t \leq e(2.0351z - 1.9047). \quad (37)$$

Indeed, $\log(1/\tau) \leq d/q$, $\log(1/x) \geq 1 - x = u \geq 1.9047e$, $1/q \leq 2.0351$, and $(m-1)d \leq ze$.

After division by $\alpha^3 p^{m-2}$, inequality (36) follows if

$$x^{m-2} \min(1, t_+) + \ell^2 y^{m-2} \leq 0.8142\alpha\kappa^2 m. \quad (38)$$

Now $0.8142\alpha \geq 0.8142 \cdot 59/120 > 0.4003$. Also $\ell^2, y^2 \leq 0.034$ and $m \geq 55$, whereas $\kappa^2 m > 0.864/m$; hence

$$\ell^2 y^{m-2} < 10^{-37} \kappa^2 m.$$

It therefore suffices to prove

$$x^{m-2} \min(1, t_+) \leq 0.39\kappa^2 m. \quad (39)$$

Since $m \geq 55$,

$$(m-2)u \geq 1.9047 \frac{m-2}{m} T \geq 1.8354T, \quad x^{m-2} \leq e^{-1.8354T}, \quad \kappa^2 m = \frac{z^2 e}{T}.$$

If $t \leq 1$, then (37) and

$$\frac{(2.0351z - 1.9047)_+}{z^2} \leq \frac{2.0351^2}{4 \cdot 1.9047} < 0.5437$$

reduce (39) to $Te^{-1.8354T} \leq 0.39/0.5437$. This holds because

$$\max_{T>0} Te^{-1.8354T} = \frac{1}{1.8354e} < 0.2005 < \frac{0.39}{0.5437}.$$

If $t > 1$, then (37) implies $z \geq 0.4753/e$. Hence the right side of (39) is at least $0.0881/(eT)$, while

$$eTe^{-1.8354T} \leq \frac{0.2005}{60} < 0.0034 < 0.0881.$$

Thus (39), and hence (36), holds. $\square$

6.2 Moderate e and the Turán corner

Lemma 6.3 (Zone C, $e \geq 1/60$). *Lemma 6.1 holds when $\frac{1}{60} \leq e < \frac{1}{3}$.*

The proof rests on three structural facts and splits at $\delta := \alpha - \frac{1}{3} = \frac{1-3e}{6}$ (note the domain constraint $q > \frac{1}{3}$ is exactly $d < \delta$): the range $\delta \geq \frac{1}{2000}$ (i.e. $e \leq \frac{1}{3} - \frac{1}{1000}$) is certified by exact finite subdivision, and the Turán-corner sliver $\delta < \frac{1}{2000}$ is Lemma 6.5 below.

Lemma 6.4 (Three-geometric defect and secant gate). *On all of $\mathcal{D}$, with $s := q/p$ (so $0 \leq y < s < x < 1$):*

$$(i) \quad \frac{R_m}{\alpha^3 p^{m-2}} = \frac{x^{m-2}}{\alpha} + \frac{\ell^2 y^{m-2}}{\alpha} - \frac{pq}{\alpha^3} s^{m-2}, \text{ and } \frac{pq}{\alpha^3} = \frac{1}{\alpha} + \frac{\ell^2}{\alpha} \text{ identically; consequently}$$

$$R_m > 0 \iff x^{m-2} - s^{m-2} > \ell^2 (s^{m-2} - y^{m-2}).$$

(ii) (*Secant gate.*) If $R_m > 0$ and $m \geq 15$, then

$$(m-2) \frac{d}{q} \geq (m-2) \ln \frac{\alpha}{q} > \ln(1+G_2), \quad G_2 := \ell^2 \left(1 - \left(\frac{y}{s}\right)^{13}\right).$$

(iii) (*Payments, m -free-gated.*) Let $\rho_{10} := (1-x^{14}) \frac{\sqrt{\alpha}}{2\sqrt{2}f(1+x)}$; then $\rho \geq \rho_{10}$, and: if $2\rho_{10}\xi \leq 1$,

$$\frac{C_m \psi(\xi, \rho)}{\alpha^3 p^{m-2}} \geq m \cdot \frac{2(1-x^{14})(1-y^{m-1}) \kappa^2 (1+4\xi)}{(1+x)(1+y)(1+2\xi)};$$

if $2\rho_{10}\xi > 1$,

$$\frac{C_m \psi(\xi, \rho)}{\alpha^3 p^{m-2}} \geq m \cdot \frac{\sqrt{2\alpha} (1-y^{m-1}) f d (4\xi+1)}{\alpha^3 (1+y) (4\xi+2)}.$$

Proof. (i) Divide $R_m = \alpha^m + L^m - pq^{m-1}$ by $\alpha^3 p^{m-2}$. The three terms become, respectively,

$$\frac{x^{m-2}}{\alpha}, \quad \frac{\ell^2 y^{m-2}}{\alpha}, \quad \frac{pq}{\alpha^3} s^{m-2},$$

and the coefficient identity is $pq - \alpha^2 = L^2$.

(ii) From (i),

$$x^{m-2} > s^{m-2} + \ell^2 s^{m-2} (1 - (y/s)^{m-2}) \geq s^{m-2} (1 + G_2),$$

using $m-2 \geq 13$ and $y/s < 1$. Hence

$$(m-2) \log(x/s) > \log(1+G_2), \quad \log(x/s) = \log(\alpha/q) \leq \frac{\alpha - q}{q} = \frac{d}{q}.$$

(iii) By the mediant inequality and $k_m(\alpha) \leq k_m(L)$,

$$\frac{A_m}{B_m} \geq \frac{k_m(\alpha)}{k_m(L)} = \frac{(p^{m-1} - \alpha^{m-1})(p+L)}{(p^{m-1} - L^{m-1})(p+\alpha)} \geq (1-x^{m-1}) \frac{p}{p+\alpha} \geq \frac{1-x^{14}}{1+x},$$

giving $\rho \geq \rho_{10}$. If $2\rho_{10}\xi \leq 1$, take $\lambda = 2\rho_{10}\xi \in [0, 1]$ in the dual form of ψ : since the dual value $\lambda\xi - \frac{\lambda^2}{4(\rho+\lambda)}$ is increasing in ρ ,

$$C_m \psi \geq C_m \left(2\rho_{10}\xi^2 - \frac{4\rho_{10}^2 \xi^2}{4(\rho_{10} + 2\rho_{10}\xi)} \right) = C_m \rho_{10} \xi^2 \frac{1+4\xi}{1+2\xi} = \frac{2\alpha^3 B_m (1-x^{14}) \kappa^2}{1+x} \cdot \frac{1+4\xi}{1+2\xi},$$

by the exact algebra $C_m \xi^2 = 4\alpha^2 B_m f \sqrt{2\alpha} d^2 / e^2$ and the definition of ρ_{10} ; then insert $B_m \geq m p^{m-2} (1-y^{m-1}) / (1+y)$. If $2\rho_{10}\xi > 1$ then $2\rho\xi > 1$, so $\xi > \xi_c(\rho)$ and $\psi = \xi - \frac{1}{4(1+\rho)} \geq \xi \frac{4\xi+1}{4\xi+2}$ (the last step is equivalent to $4\xi\rho \geq 2$); multiply by C_m , use $C_m \xi = \sqrt{2\alpha} B_m f d$ and the same B_m bound. $\square$

Proof of Lemma 6.3 for $e \leq \frac{1}{3} - \frac{1}{1000}$. By Lemma 6.4, it suffices that for every box of a finite cover of $\{(e, \kappa)\}$ and every $m \geq 15$, the (upper-bounded) three-geometric defect not exceed m times the (lower-bounded) applicable payment coefficient; positive defect only needs checking for $m \geq M_+ := 3 + \lfloor q(G_2 - G_2^2/2)/d \rfloor$ (the strict secant gate with $\ln(1+t) \geq t - t^2/2$), and all sufficiently large m pass at once because both positive defect terms decrease geometrically while the payment grows linearly. The script `zoneC_certifier.py` certifies exactly this, in exact rational arithmetic, over $e \in [\frac{1}{60}, \frac{1}{3} - \frac{1}{1000}]$, $\kappa \in (0, 1]$: 2997 boxes verified (of which 5 by the analytic $\kappa \rightarrow 0$ bottom-out,

which combines the gate $m - 2 > C_0/\kappa$ with the monotonicity of $e^{-a/\kappa}/\kappa$, maximum subdivision depth 28, largest explicitly checked cycle length $m = 1716$. The interval construction uses the lower endpoint

$$f_{\text{lo}} = \max\{\alpha_{\text{lo}} - L_{\text{up}}, d_{\text{lo}} + \delta_{\text{lo}}\},$$

whose validity is exactly Lemma 3.4; this is the bound that was missing from the supplied manuscript. $\square$

6.3 The Turán-corner sliver

Lemma 6.5 (Turán corner). *Lemma 6.1 holds when $0 < \delta = \frac{1-3e}{6} < \frac{1}{2000}$ (equivalently $e > \frac{1}{3} - \frac{1}{1000}$).*

Proof. Write

$$\alpha = \frac{1}{3} + \delta, \quad e = \frac{1}{3} - 2\delta, \quad q = \frac{1}{3} + \delta - d, \quad p = \frac{2}{3} - \delta + d,$$

where $0 < d < \delta \leq 1/2000$. Since

$$\frac{\alpha}{e} \leq \frac{2003}{1994}, \quad \xi = 4 \left(\frac{\alpha}{e} \right)^2 d < 4.04 d < 1,$$

we are indeed in Zone C.

We first collect bounds that will be used repeatedly. Put $S := q^2 - L^2 = (q - L)(q + L)$. Direct expansion gives

$$S = \delta + 3\delta^2 - \left(\frac{1}{3} + 4\delta \right) d + 2d^2. \quad (40)$$

Writing $d = t\delta$, $0 \leq t \leq 1$, shows

$$\frac{S}{\delta} = 1 - \frac{t}{3} + \delta(3 - 4t + 2t^2), \quad \frac{2}{3} \leq \frac{S}{\delta} \leq 1 + 3\delta \leq \frac{2003}{2000}.$$

Moreover

$$L^2 - e^2 = \delta - \frac{d}{3} - 6\delta^2 + 2\delta d - d^2 \geq \frac{2}{3}\delta - 5\delta^2 > 0,$$

because the displayed expression is decreasing in $d \in [0, \delta]$. Thus $e < L < q$, and consequently

$$\frac{1997}{3000} \leq q + L \leq \frac{2003}{3000}. \quad (41)$$

Also

$$\frac{1}{2} \leq x = \frac{\alpha}{p} \leq \frac{1}{2} + 2.26\delta \leq 0.50113, \quad y < \frac{1}{2}, \quad x^{14} < 6.4 \cdot 10^{-5}.$$

Here $2\alpha - p = 3\delta - d > 0$, while $x - \frac{1}{2} = (3\delta - d)/(2p) \leq 2.26\delta$.

The identities

$$q - L = \frac{S}{q + L}, \quad f = \alpha - L = \frac{\alpha^2 - L^2}{\alpha + L} = \frac{3\alpha\delta + d(d + e)}{\alpha + L}$$

and (40)–(41) imply

$$\frac{2000}{2003} \delta \leq q - L \leq 1.505 \delta, \quad \frac{3000}{2003} \delta \leq f \leq 2.005 \delta, \quad 1 - \ell = \frac{f}{\alpha} \leq 6.02 \delta. \quad (42)$$

For the upper bound on f we used $L \geq e$, and hence $\alpha + L \geq 2/3 - \delta$.

Step 1: positive defect forces a visible frontier gap. Set

$$t_0 := 1 - \frac{y}{s} = \frac{q - L}{q}.$$

The preceding estimates give

$$2.99\delta \leq t_0 \leq 4.52\delta, \quad \ell^2 \geq 1 - 12.1\delta.$$

Taylor's formula with nonnegative third-order remainder gives $1 - (1 - t)^{13} \geq 13t - 78t^2$ for $0 \leq t \leq 1$. Therefore

$$G_2 = \ell^2(1 - (1 - t_0)^{13}) \geq (1 - 12.1\delta)(13 \cdot 2.99\delta - 78(4.52\delta)^2) > 37.8\delta.$$

(The quotient of the middle expression by δ decreases on $[0, 1/2000]$, and its endpoint value is greater than 37.84.) On the other hand $G_2 \leq 13t_0 \leq 58.8\delta \leq 0.0294$. Hence

$$\log(1 + G_2) \geq G_2 \left(1 - \frac{G_2}{2}\right) > 37.2\delta.$$

Lemma 6.4(ii), together with $q > 1/3$, now yields

$$R_m > 0 \implies (m - 2)d > 12.4\delta. \quad (43)$$

Step 2: an upper bound on the normalized defect. Put $n = m - 2$. From Lemma 6.4(i), the mean-value theorem on $[s, x]$ and $[y, s]$, and $y = \ell x$, we obtain

$$\frac{\alpha R_m}{\alpha^3 p^{m-2}} \leq n(x - s)x^{n-1} - \ell^2 n(s - y)y^{n-1} = \frac{nx^{m-3}}{p} N,$$

where

$$N := d - \frac{S}{q + L} \ell^{m-1}.$$

Using (40), dropping the nonpositive $-2d^2\ell^{m-1}/(q + L)$ term, using $\ell^{m-1} \leq 1$ in the positive d term, and using

$$\ell^{m-1} \geq 1 - (m - 1)(1 - \ell) \geq 1 - 6.02(m - 1)\delta$$

in the negative δ term, gives

$$N \leq d \left(1 + \frac{2\alpha - e}{q + L}\right) - \frac{3\alpha\delta}{q + L} + \frac{3\alpha \cdot 6.02(m - 1)\delta^2}{q + L}.$$

Since $d \leq \delta$,

$$d \left(1 + \frac{2\alpha - e}{q + L}\right) - \frac{3\alpha\delta}{q + L} \leq \delta \frac{L - e - d}{q + L} \leq \frac{3\delta^2}{q + L} < 4.51\delta^2.$$

The last term is less than $9.06m\delta^2$, by $\alpha \leq 2003/6000$ and (41). Thus $N < 9.5m\delta^2$. Finally $(p\alpha)^{-1} < 4.51$, and therefore

$$\frac{R_m}{\alpha^3 p^{m-2}} < 43m(m - 2)\delta^2 x^{m-3}. \quad (44)$$

Step 3: the two payments. Lemma 6.4(iii), together with $\kappa = d/e \geq 3d$, $y < 1/2$, (42), and the bounds above, gives

$$2\rho_{\text{lo}}\xi \leq 1 \implies \frac{C_m\psi(\xi, \rho)}{\alpha^3 p^{m-2}} > 7.9 d^2 m, \quad (45)$$

$$2\rho_{\text{lo}}\xi > 1 \implies \frac{C_m\psi(\xi, \rho)}{\alpha^3 p^{m-2}} > 10.9 \delta d m. \quad (46)$$

For completeness, the first numerical coefficient follows from

$$\frac{2(1 - 6.4 \cdot 10^{-5})(1 - 2^{-14})9}{1.50113 \cdot 1.5} > 7.9.$$

For the second, use $\sqrt{2\alpha} > 0.816$, $f > 1.497\delta$, $\alpha^3 < 0.0373$, $(4\xi + 1)/(4\xi + 2) \geq 1/2$, and

$$\frac{0.816(1 - 2^{-14})1.497}{2 \cdot 0.0373 \cdot 1.5} > 10.9.$$

Step 4: comparison. Assume $R_m > 0$. In the first branch, (43) and (45) reduce the desired comparison with (44) to

$$43(m-2)^3 x^{m-3} < 7.9 \cdot 12.4^2.$$

In the second branch, (46) reduces it to

$$43(m-2)^2 x^{m-3} < 10.9 \cdot 12.4.$$

For $j = 2, 3$, the sequence $(m-2)^j x^{m-3}$ decreases for $m \geq 15$, since

$$x \left(\frac{m-1}{m-2} \right)^3 \leq 0.50113 \left(\frac{14}{13} \right)^3 < 1.$$

It therefore suffices to check $m = 15$, where

$$43 \cdot 13^3 \cdot 0.50113^{12} < 24 < 7.9 \cdot 12.4^2, \quad 43 \cdot 13^2 \cdot 0.50113^{12} < 1.83 < 10.9 \cdot 12.4.$$

Thus the applicable payment dominates the defect in either branch. $\square$

Proof of Lemma 6.3. Combine the certified range $e \leq \frac{1}{3} - \frac{1}{1000}$ with Lemma 6.5. $\square$

7 Assembly

Proof of Theorem 1.1. Let $(q, \alpha, m) \in \mathcal{D}$. If $R_m \leq 0$ there is nothing to prove, since $\psi \geq 0$ and $C_m > 0$ ($B_m > 0$ always, and $f > 0$ because $L^2 \leq \alpha e < \alpha^2$: indeed $e < \frac{1}{3} < \frac{1-e}{2} = \alpha$). So assume $R_m > 0$. If $\xi \leq 1$, apply Lemma 6.1. If $\xi \geq 1$ and $e \leq \frac{1}{60}$, apply Lemma 4.1; if $\xi \geq 1$ and $e \geq \frac{1}{60}$, apply Lemma 5.1. In each case the certified quantity is a value of the dual functional $\lambda \mapsto C_m(\lambda\xi - \frac{\lambda^2}{4(\rho+\lambda)})$ at an admissible $\lambda \in [0, 1]$, hence $\leq C_m\psi(\xi, \rho)$ by Proposition 2.12. $\square$

Proof of Corollary 1.2. Let W have density $p \in (\frac{1}{2}, \frac{2}{3})$ and let $m \geq 3$ be odd. For $m \leq 13$ the conjecture is proved in `earlier_consolidated_note.tex` at all densities. Let $m \geq 15$. If $\lambda_{\max}(A) \leq q$, the bound follows from Lemma 2.2. Otherwise, by Lemmas 2.3 and 2.4, A has a unique frontier eigenvalue $\alpha \in (q, r(q)]$, so $(q, \alpha, m) \in \mathcal{D}$ and Theorem 1.1 gives $R_m \leq C_m\psi(\xi, \rho)$. By Theorem 2.11 and Proposition 2.7,

$$t(C_m, W) - (p^m - pq^{m-1}) \geq -R_m + A_m c^2 + B_m \|g_s\|_2^2 \geq -R_m + C_m\psi(\xi, \rho) \geq 0. \quad \square$$

Proof of Corollary 1.3. For $p \leq \frac{1}{2}$ the right-hand side of (1) is nonpositive. For $p \geq \frac{2}{3}$ this is the high-density theorem of `earlier_consolidated_note.tex`. For $\frac{1}{2} < p < \frac{2}{3}$ this is Corollary 1.2. $\square$

8 Exact certificate protocol

This section records precisely what the two finite verifiers do. It is part of the proof specification: a replacement implementation is valid if it checks the same interval inequalities with outward rounding.

The programs certify the relative interiors of their strips. A box that lies entirely on an overlap or an excluded boundary may be discarded by a one-sided domain test: the overlap $\xi = 1$ is already covered by Zone C, the boundary $q = 1/3$ is outside the stated domain, and the admissible ceiling $\alpha = r(q)$ follows from the certified interior by continuity of all scalar quantities. Thus no admissible endpoint is lost by the strict skip tests.

8.1 Arithmetic primitives

Both certifiers use Python’s `fractions.Fraction`; hence every field operation and every comparison is exact. Square roots are enclosed by integer square roots: for a fixed decimal scale S ,

$$\frac{\lfloor \sqrt{\lfloor tS^2 \rfloor} \rfloor}{S} \leq \sqrt{t} \leq \frac{\lfloor \sqrt{\lfloor tS^2 \rfloor} \rfloor + 1}{S}.$$

For $t \geq 0$, an upper bound on e^{-t} is obtained by first rounding t down rationally and then using

$$e^{-t} \leq \left(\sum_{j=0}^N \frac{t^j}{j!} \right)^{-1}.$$

Where a lower bound is requested by the diagnostic checker, the omitted positive tail of the exponential series is bounded by a geometric tail. Integer powers in the Zone-C verifier are evaluated by binary powering with a directed rational rounding after every multiplication. The only floating-point conversions in the exact certifiers occur in error messages or printed summaries; no branch that returns “verified” depends on them.

8.2 Zone B

The verifier starts from

$$\frac{1}{60} \leq e \leq \frac{2033}{10000}, \quad \kappa_\xi(e) \leq \kappa \leq \min\{\kappa_{\max}(e), \kappa_q(e)\}.$$

A rational rectangle is discarded only if monotonic endpoint tests show that it lies wholly outside this strip. On every remaining rectangle the program forms outward bounds for $x, A, \lambda, \bar{\ell}, \rho, \varepsilon, \Phi$, the endpoint value $K(14)$, and the interior maximum bound $e^{-\Phi-2w(\Phi)}$. It then checks exactly the box version of (34). A failed box is bisected in the coordinate with the larger relative width. The run terminates with 23 verified leaves, 3 discarded leaves and maximum depth 8; reaching the depth cap with an unverified box causes a nonzero exit.

8.3 Zone C on the compact middle range

Here the initial domain is

$$\frac{1}{60} \leq e \leq \frac{1}{3} - \frac{1}{1000}, \quad 0 < \kappa \leq \min\{\kappa_\xi(e), \kappa_{\max}(e), \kappa_q(e)\}.$$

For each rational box the verifier encloses $\alpha, d, q, p, x, s, L, y, \ell, \xi$ and the two payment coefficients in Lemma 6.4(iii). In particular it uses

$$f_{\text{lo}} = \max\{\alpha_{\text{lo}} - L_{\text{up}}, d_{\text{lo}} + \delta_{\text{lo}}\},$$

which is valid by Lemma 3.4. The secant gate says that positive defect is possible only for

$$m \geq 3 + \left\lfloor \frac{q_{\text{lo}}(G_{2,\text{lo}} - G_{2,\text{lo}}^2/2)}{d_{\text{up}}} \right\rfloor.$$

Starting there, the program checks each integer m with an outward upper bound on the three-geometric defect. Once the sum of the two positive geometric terms is below the linearly growing payment, all later m are certified at once. Boxes touching $\kappa = 0$ are closed by the analytic bottom-out estimate

$$m - 2 > \frac{C_0}{\kappa}, \quad \frac{R_m}{\alpha^3 p^{m-2}} \leq \frac{1}{\alpha_{\text{lo}}} e^{-a/\kappa},$$

together with monotonicity of $e^{-a/\kappa}/\kappa$ on the verified range. The completed run has 2997 verified leaves, including 5 bottom-out leaves, 87 discarded leaves, maximum depth 28, and largest explicitly tested exponent $m = 1716$. As in Zone B, any unresolved leaf causes a nonzero exit.

9 Verification

Following the standing practice of this project, every claim above carries one of three labels, and the accompanying scripts let a reader re-check each label mechanically.

- (1) *Analytic.* The one-frontier reduction, including the no-frontier case (§2), the coordinate and defect identities (§3), Zone A, the m -maximization Lemma 5.2, the Zone-C small- e lemma, the Turán-corner lemma, and the assembly are complete pencil-and-paper proofs. Every terminating decimal used in those arguments is checked as an exact rational inequality by the first layer of the companion checker.
- (2) *Exact certificate.* The Zone-B battle (Lemma 5.3) and the moderate- e part of Zone C (Lemma 6.3, range $e \leq \frac{1}{3} - \frac{1}{1000}$) are proved by finite exhaustive box subdivisions in *exact rational arithmetic* (`zoneB_certifier.py`: 23 boxes; `zoneC_certifier.py`: 2997 boxes with a per-box finite m -scan plus gate and tail lemmas); no floating point enters the certified statements. The transcendentals $\sqrt{t}$ and e^{-t} are bracketed by integer square roots and by exponential-series partial sums with explicit tail bounds.
- (3) *Numerical (safety net only; no theorem depends on it).* `regionII_scalar_checker.py` first checks the algebraic identities and all directed constants exactly (using `Fraction` and `SymPy`), then tests every analytic chain on adversarial high-precision grids. With `--scan`, the recorded reproducible run examined 1359 positive-defect parameter triples, through $m = 1,000,001$ on the prescribed grid. It found no target violation and no point missed by both dual certificates; the smallest relative payment margin in that run was $4.638 \cdot 10^{-5}$, at $(e, \kappa, m) = (10^{-10}, 0.5999999998, 100001)$. These data are useful for transcription control only and are not invoked by any proof.

How to audit

- (a) Audit §2 independently from the block decomposition onward. The points most easily missed are Remarks 2.5, 2.6, the no-frontier Lemma 2.2, and the $K = 0$ case in Theorem 2.11.
- (b) Check the chart Lemma 3.2 by direct computation (or run the exact identity layer of the checker).
- (c) Zone A: the only nontrivial verifications are the elementary inequality (18), the constants in (22), and the three ranges of the battle (31); each range is a one-variable calculus argument spelled out in full.
- (d) Zone B: verify Lemma 5.2 (five lines of calculus), then run `zoneB_certifier.py`.
- (e) Zone C: the small- e case reuses the Zone-A machinery plus the battle (39); the moderate- e case is Lemma 6.4 (a page of identities) plus `zoneC_certifier.py`; the Turán-corner Lemma 6.5 is self-contained, and every displayed constant of its proof is re-verified as an exact rational comparison by the checker.
- (f) Run `python3 regionII_scalar_checker.py --scan` for the numerical safety net.